# Brainware University

## Department of Computer Science & Engineering - Artificial Intelligence

## Final Year Project Progress Report

# Project Title: Soil Health Analysis and Crop Suitability Prediction Using ML


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**Date of Submission:**

# INTRODUCTION

In today's digital world, Agriculture plays a critical role in ensuring food security and sustainable economic development. Soil health is one of the most important factors influencing crop productivity and long-term agricultural sustainability. However, traditional soil analysis and crop selection methods are often manual, time-consuming, and may not fully utilize available data.

The proposed project, *"Soil Health Analysis and Crop Suitability Prediction Using Machine Learning,"* aims to apply data-driven techniques to analyze soil parameters and predict the most suitable crops for cultivation. By integrating soil data, environmental factors, and machine learning algorithms, the system seeks to support informed agricultural decision-making.

This first project review summarizes the progress achieved so far, representing approximately 30% of the total project work.

# OBJECTIVE

The primary objectives of the project are:

1. To collect and preprocess soil health and environmental datasets.
2. To analyze key soil parameters such as pH, Nitrogen (N), Phosphorus (P), Potassium (K), organic carbon, moisture, and electrical conductivity.
3. To develop machine learning models capable of predicting suitable crops based on soil conditions.
4. To evaluate model performance using appropriate metrics.
5. To design a basic prototype interface for demonstrating crop suitability predictions.

# PROJECT SCOPE

**In Scope:**

- Collection of soil sample datasets (public or institutional sources).
- Data preprocessing and exploratory data analysis (EDA).
- Feature engineering from soil and environmental parameters.
- Development of baseline machine learning models (e.g., Decision Tree, Random Forest, Logistic Regression).
- Model evaluation and comparison.
- Development of a basic demonstration interface.

**Out of Scope (for current phase):**

- Real-time IoT sensor integration.
- Large-scale field deployment.
- Commercial production implementation.

# WORK COMPLETED

The following tasks have been completed so far:

**4.1 Project Planning and Literature Review**

- Defined project problem statement and objectives.
- Conducted literature review on soil health indicators and crop prediction systems.

- Studied existing machine learning techniques used in agricultural applications.

### 4.2 Requirement Analysis

- Identified key soil parameters required for analysis.
- Determined dataset requirements and potential data sources.
- Selected tools and technologies (Python, Pandas, Scikit-learn, Matplotlib).

### 4.3 Data Collection (Initial Phase)

- Collected a preliminary dataset from public agricultural sources.
- Organized soil and crop data into structured format (CSV files).

### 4.4 Data Preprocessing (Partial)

- Performed initial data cleaning.
- Handled missing values (basic level).
- Standardized column names and formats.

### 4.5 Exploratory Data Analysis (Preliminary)

- Generated summary statistics.
- Created basic visualizations for soil parameters.
- Observed correlations between soil nutrients and crop types.

## 5. Work in Progress

- Advanced data preprocessing and normalization.
- Feature selection and feature engineering.
- Dataset expansion to improve model robustness.
- Implementation of baseline machine learning models.
- Defining evaluation metrics and validation strategy.

## 6. Next Steps

The upcoming tasks include:

1. Finalizing cleaned and structured dataset.
2. Implementing and training multiple machine learning models.
3. Comparing model performance using accuracy, precision, recall, and F1-score.
4. Optimizing the best-performing model.

5. Developing a simple user interface for prediction demonstration.
6. Preparing mid-term presentation and technical documentation.

## 7. Challenges Encountered

- Limited availability of high-quality and region-specific soil datasets.
- Presence of missing or inconsistent data values.
- Class imbalance in crop categories.
- Defining accurate crop suitability labels.

### Mitigation Measures

- Applying data cleaning and imputation techniques.
- Using balanced sampling methods where necessary.
- Consulting agricultural guidelines to improve labeling logic.

## 8. Project Status

The project is progressing as per the planned schedule. The foundational work including planning, literature review, data collection, and preliminary analysis has been completed. Model development and system implementation are currently underway.

## 9. Conclusion

The first phase of the project has successfully established the conceptual and technical foundation required for further development. With approximately 30% of the work completed, the project is now transitioning from data preparation to model development and evaluation.

Continued progress in dataset refinement and model implementation will enable the successful completion of the remaining phases of the project.